

	1	2	3	4	5
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch		Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch		
B					
C					
D					Sheet: / File: Audio_Visualiser_DSP.kicad_sch Title: Size: A Date: 2023-12-25 Rev: KiCad E.D.A. 10.0.3 Id: 1/20
	1	2	3	4	5

TRY TO TALLY LOAD CURRENTS

Component List:

- N201: +12V 1A Power Supply
- D201: 24V DNP
- J201: +12V IN
- TP201: GND
- TP202: +12Vin
- TP203: +12V
- PWR_FLAG: +12V
- U201: TPS26600RHFT
- R201: 55.1k 1% 1/16W
- R202: 1.15k 1% 1/16W
- R203: 5.11k 1% 1/16W
- R204: 10k 1% 1/16W
- R205: 10k 1% 1/16W
- R206: 10k 1% 1/16W
- R207: 0.02 1% 1/16W
- R208: 10k 1% 1/16W
- R209: 3.65k 1% 1/16W
- R210: 10k 1% 1/16W
- R211: 10k 1% 1/16W
- C201: 10nF 50V X7R
- C202: 0.1uF 50V X7R
- C203: 100uF 25V 10%

Pin Connections:

- IN (8) to +12Vin
- IN (9) to +12Vin
- SHDN (14) to +12Vin
- UVLO (10) to +12Vin
- OVP (12) to +12Vin
- dVdT (20) to +12Vin
- MODE (13) to +12Vin
- FLT (22) to +12Vin
- RTN (15) to GND
- RTN_PAD (25) to GND
- IMON (18) to +12Vin
- ILIM (19) to +12Vin

Notes:

- UVLO threshold set to 10.8V
- OVLO threshold set to 13.2V
- MODE set to Circuit Breaker with Auto-Retry
- Output Current Limit set to 1.2A
- +3.58V output at +13.2Vin
- +3.21V output at +12.0Vin

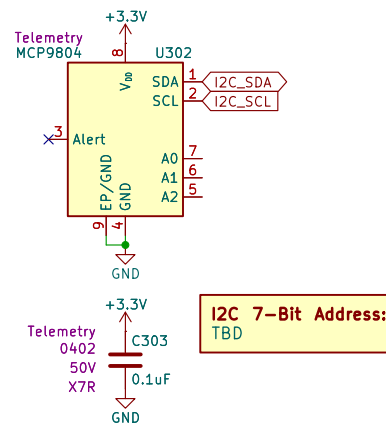
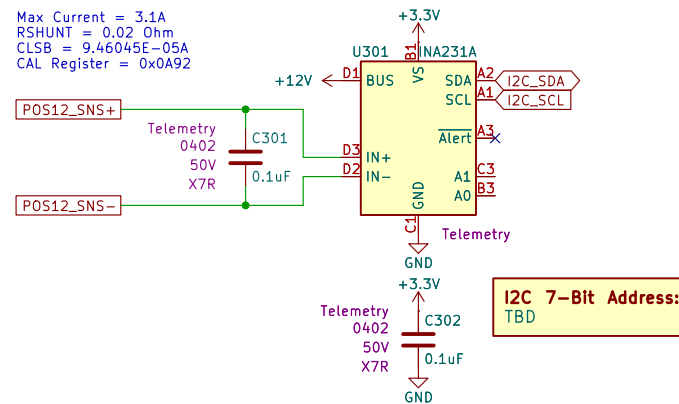
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Size: A Date: 2023-12-25 Rev: KICad E.D.A. 10.0.3 Id: 2/20

Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20

Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20



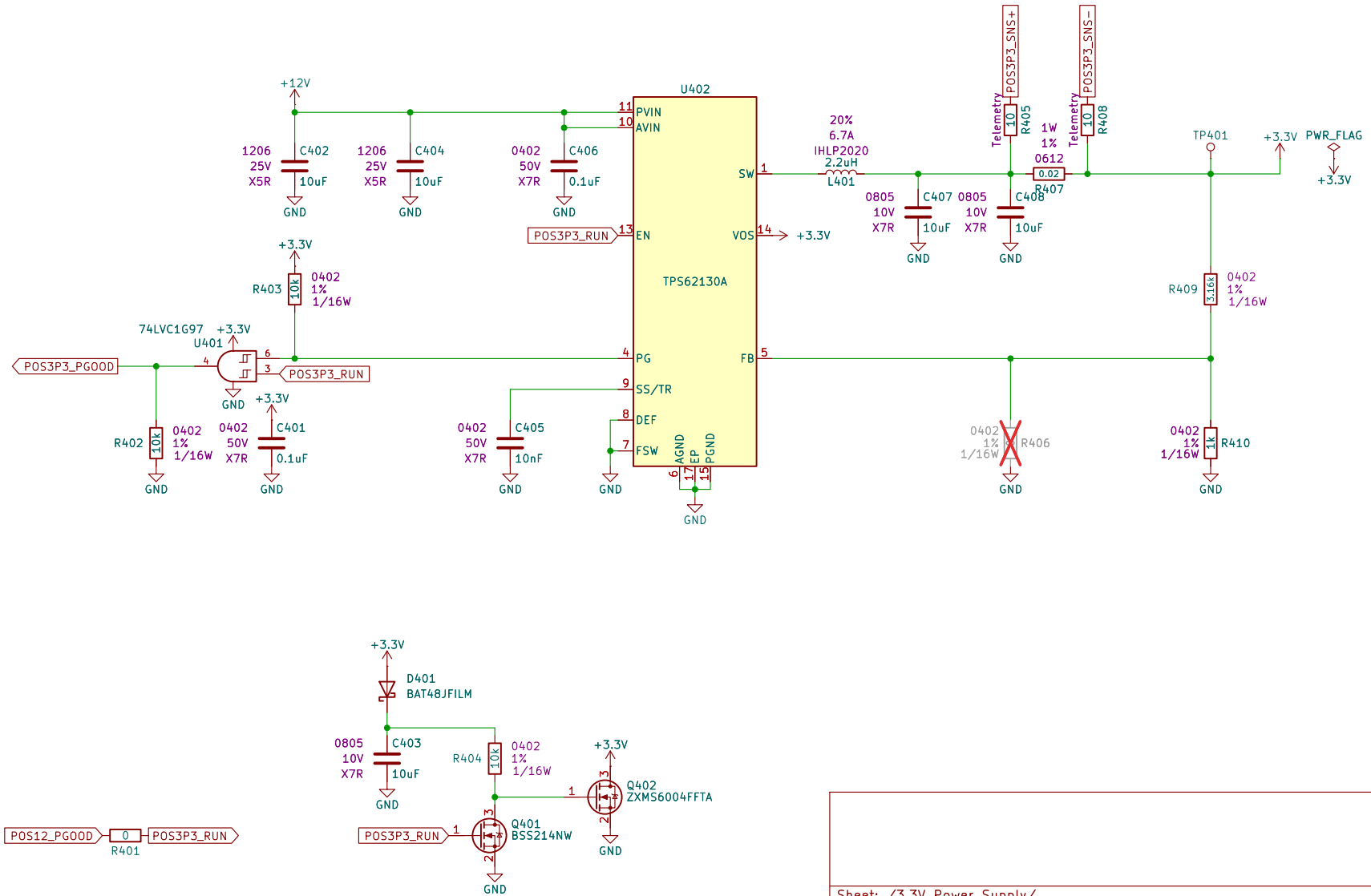
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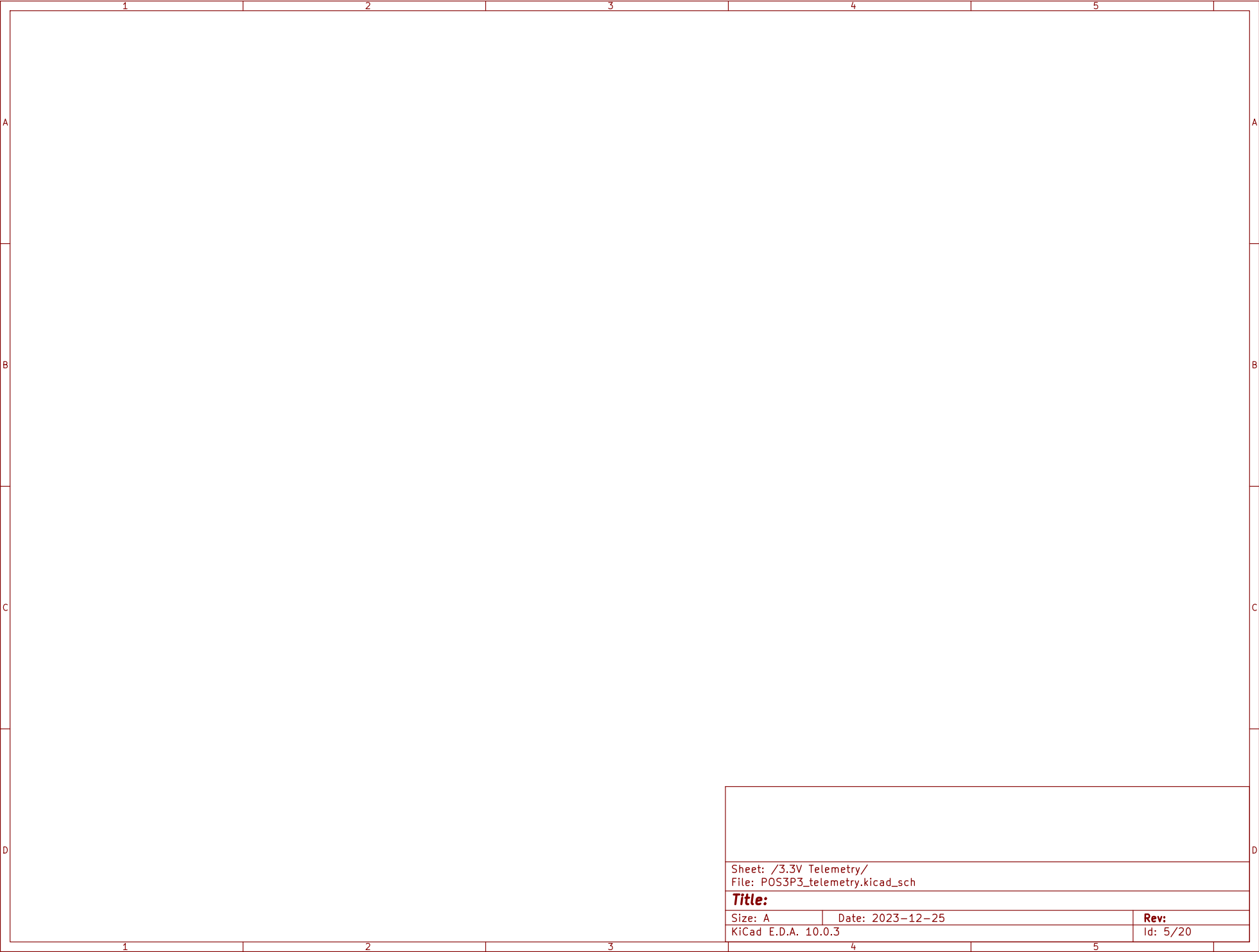
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 KiCad E.D.A. 10.0.3

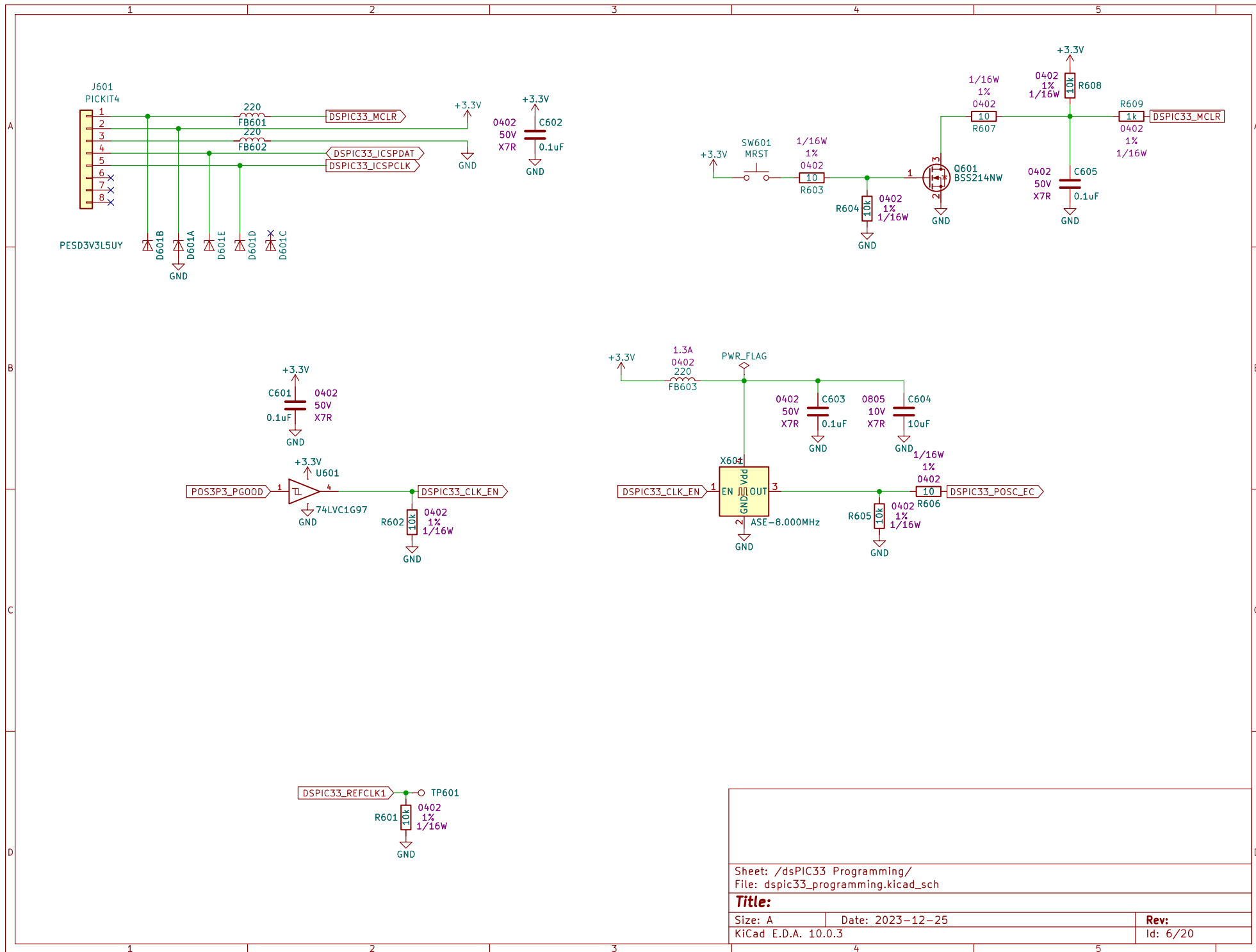
Rev:
 Id: 3/20

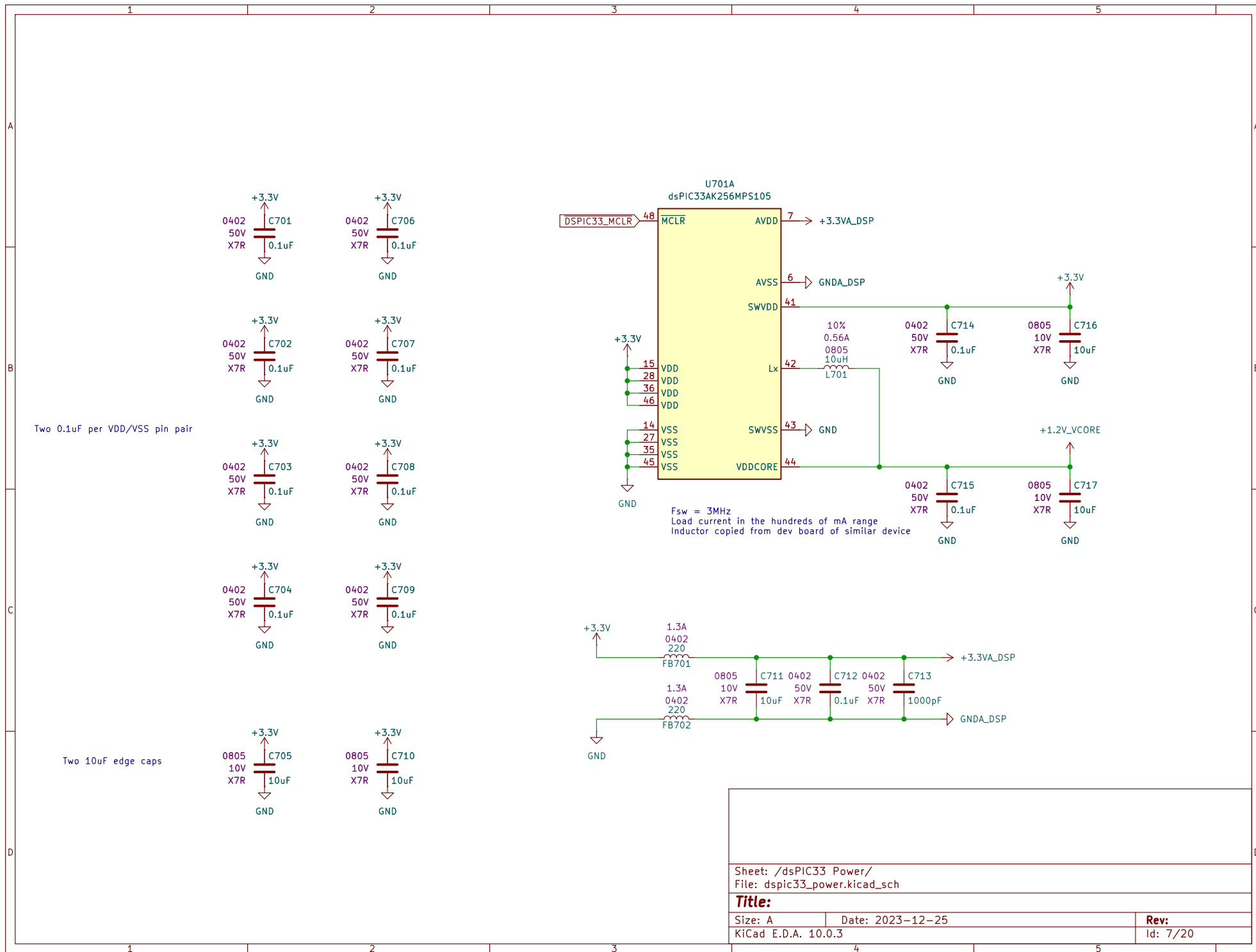
TRY TO TALLY LOAD CURRENTS

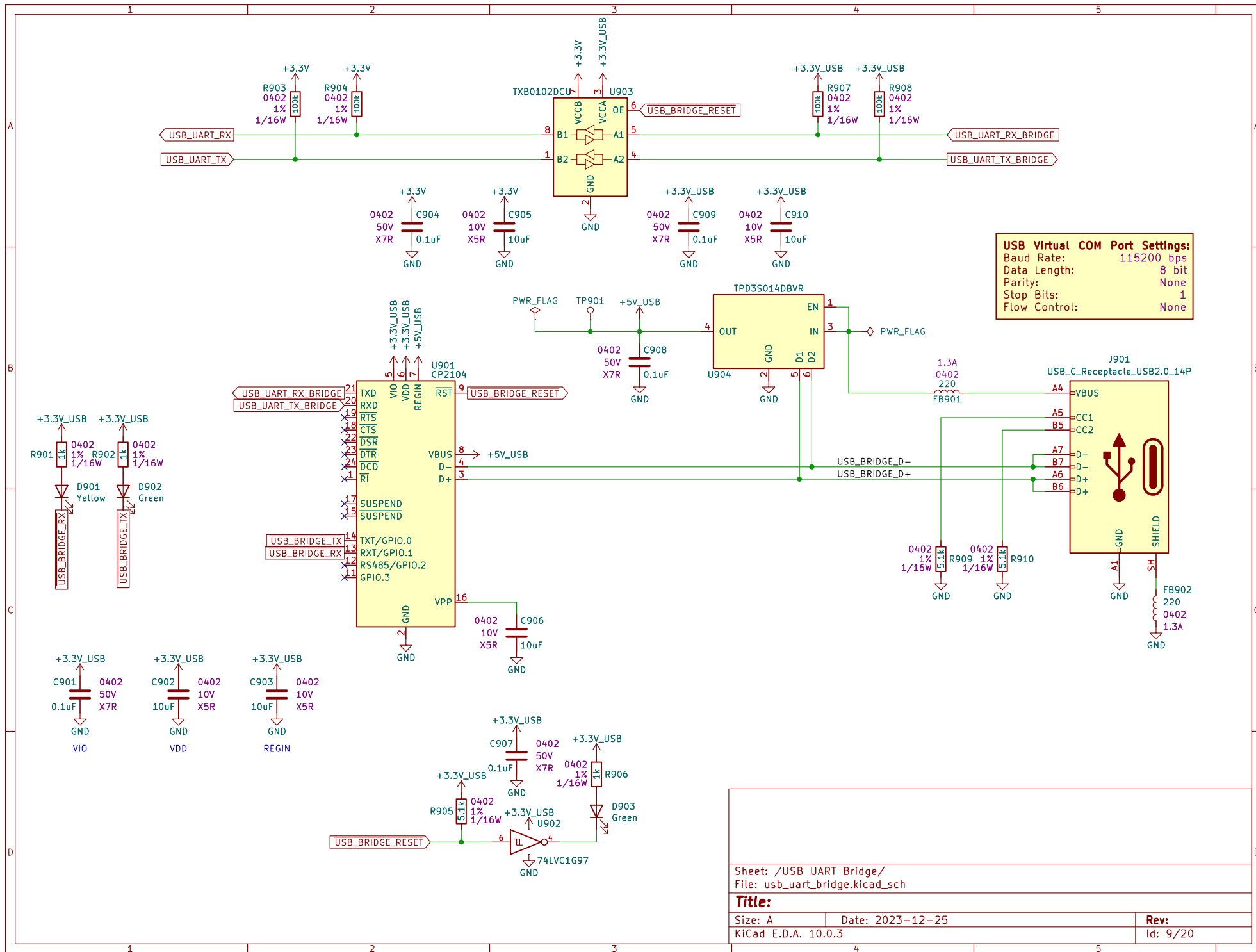


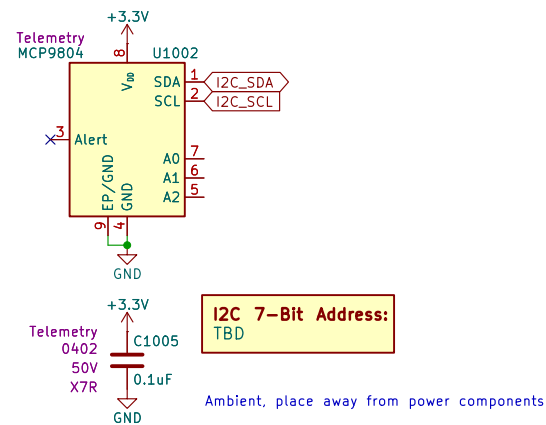
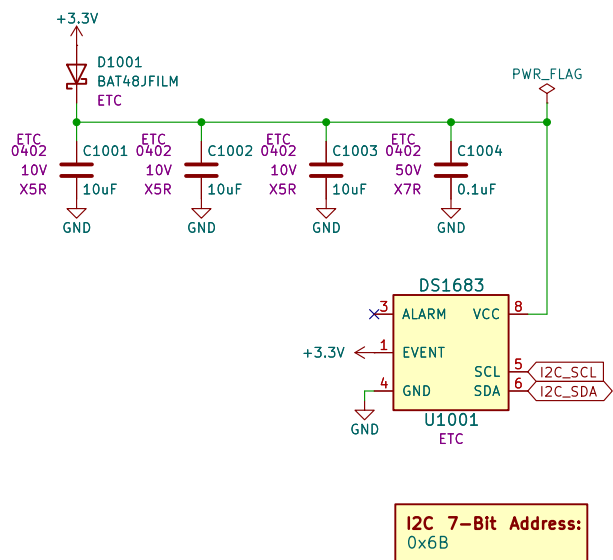
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Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 4/20









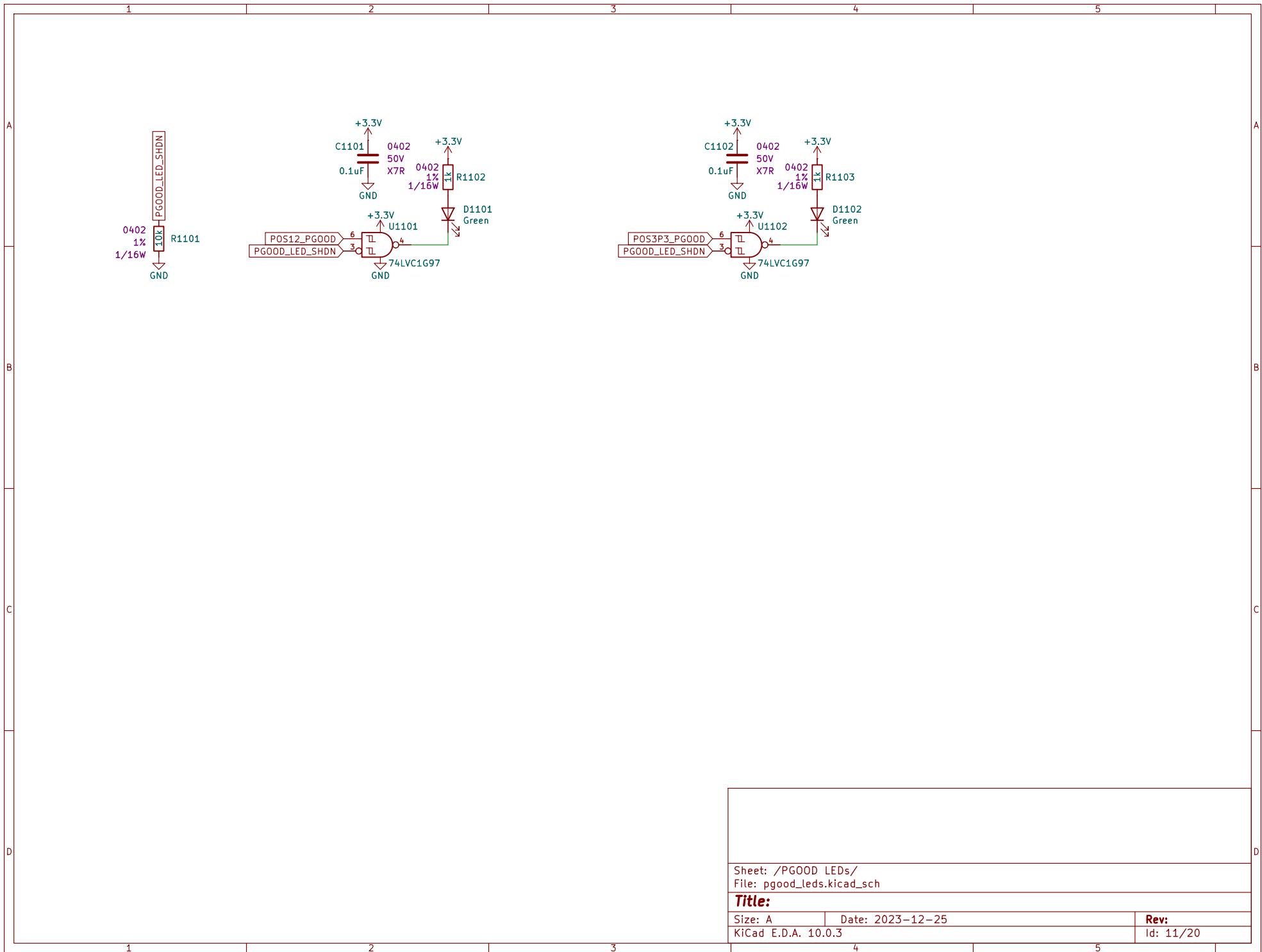


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File: platform_etc.kicad_sch

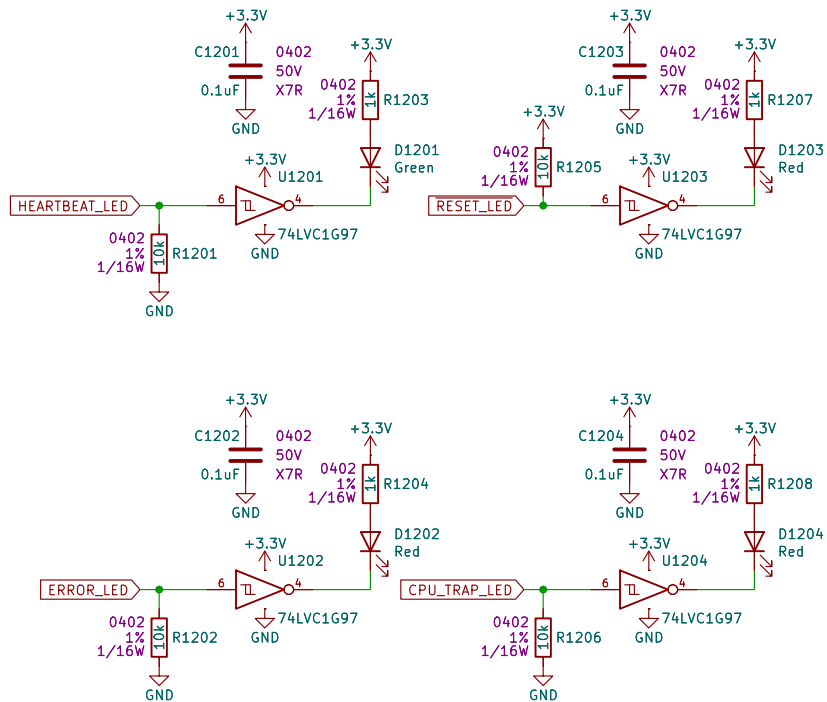
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Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

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Sheet: /PGOOD LEDs/ File: pgood_leds.kicad_sch		
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Size: A	Date: 2023-12-25	Rev:
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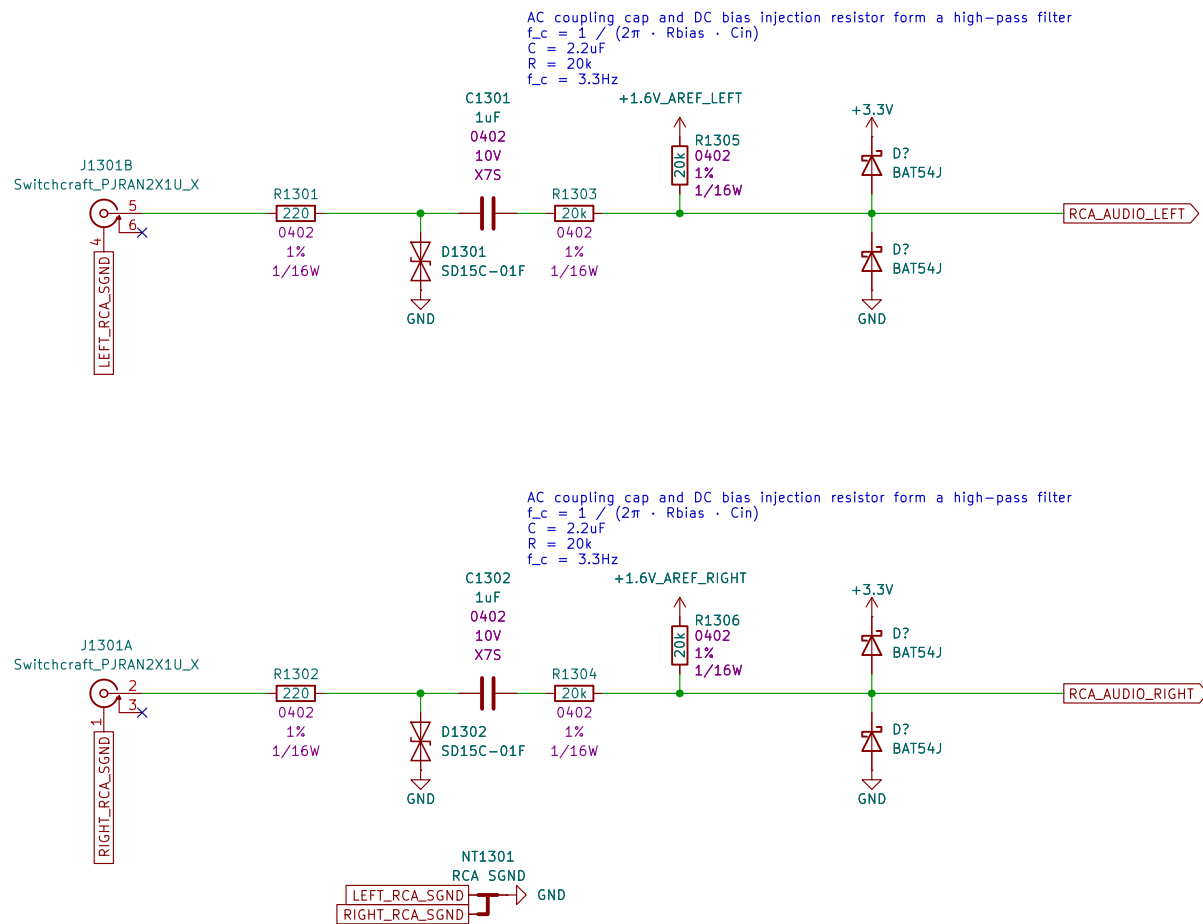


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Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/20



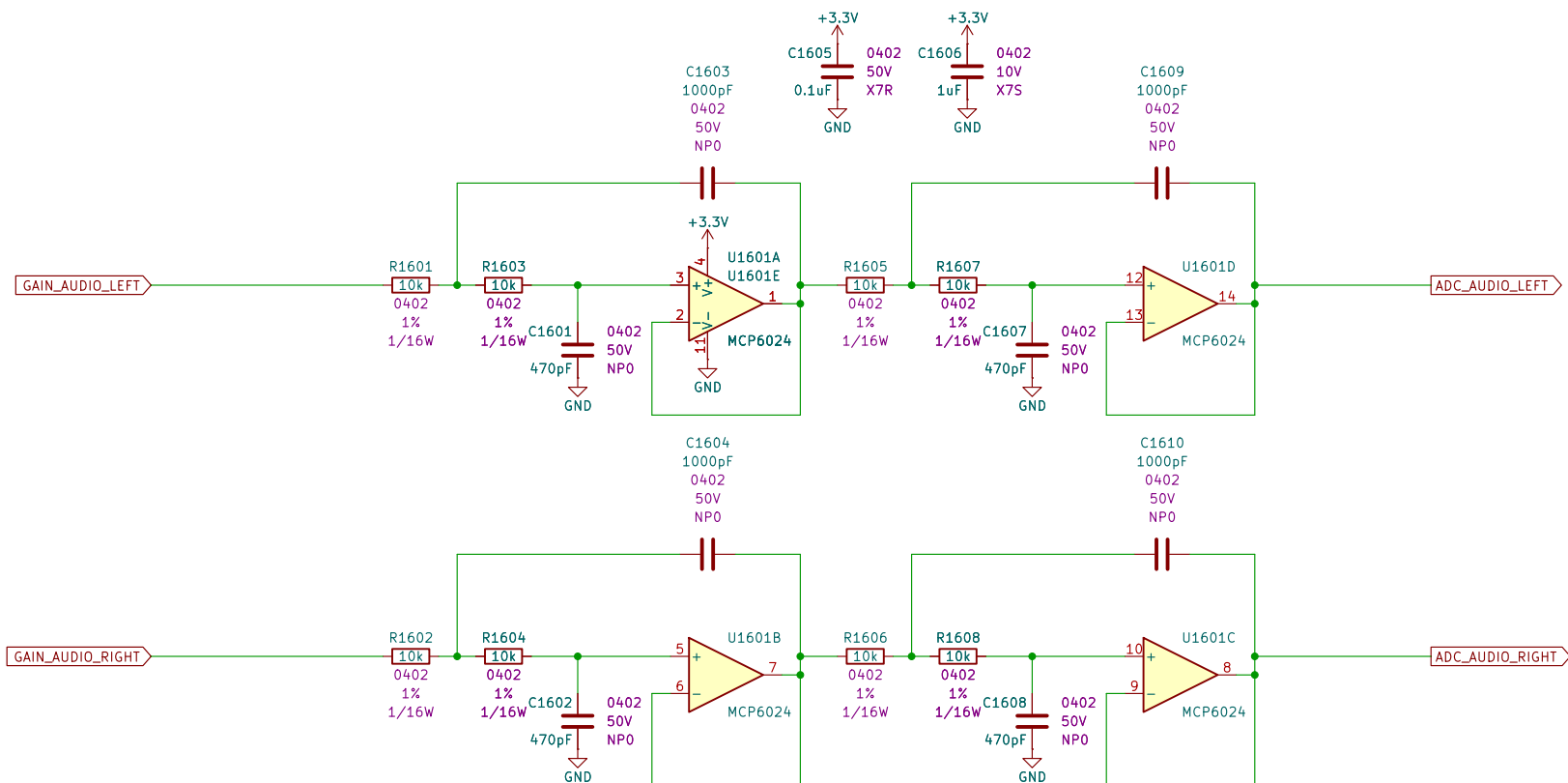
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Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

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Id: 13/20

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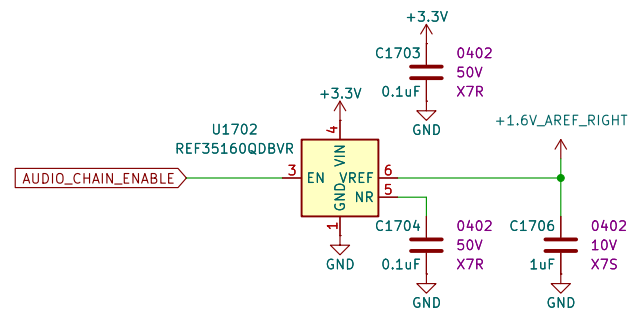
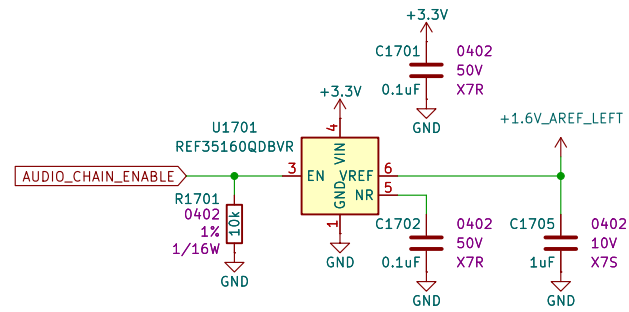
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz, nyquist is 22.05kHz
 Must filter hard around this frequency

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/20

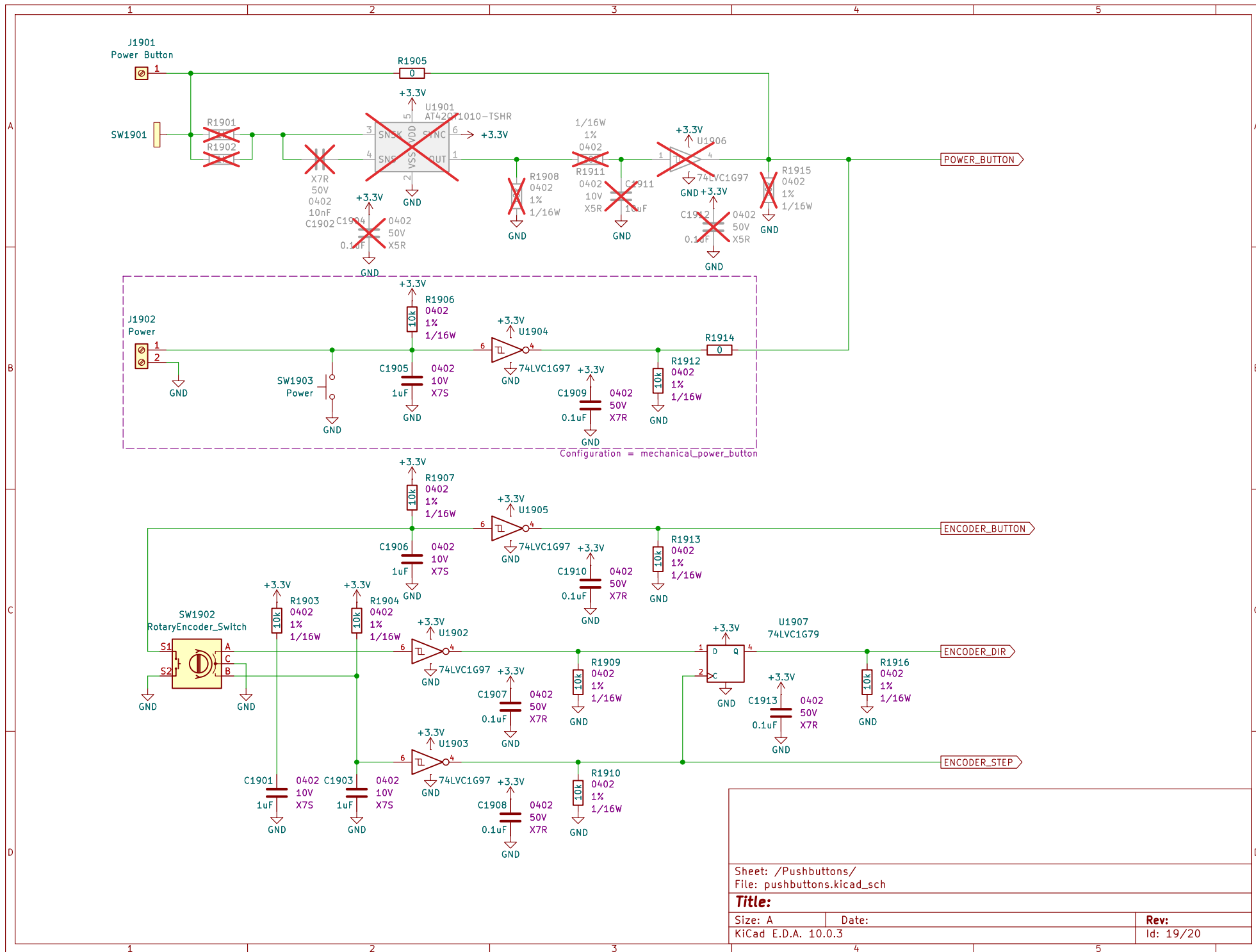


Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

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Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A Date:

KiCad E.D.A. 10.0.3

Rev:

Id: 19/20

